

Emran Lotfi

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Profile

Computer Science student with hands-on experience building and deploying full-stack software systems. Developed and shipped projects spanning AI-powered web applications, computer vision classifiers, and NLP pipelines — from system design through API development, containerization, and cloud deployment. Proficient across the stack in Python, Java, TypeScript, and SQL, with a systematic approach to engineering, testing, and documentation.

Technical Projects

DermaScan — Skin Lesion Classifier | *Python, PyTorch, FastAPI, Docker*

[Live Demo](#) | [GitHub Repo](#)

- Built a 7-class skin lesion classification system using DermaMNIST, training a custom CNN baseline and fine-tuning ResNet-18 with transfer learning to achieve 83% test accuracy.
- Ran 5 systematic experiments tracked with MLflow, comparing architectures (CNN vs. ResNet-18) and learning rates to identify that conservative fine-tuning (LR=1e-4) outperforms aggressive updates.
- Deployed as a production-ready web application with a FastAPI backend, Dockerized the full stack, and hosted on Render with a user-facing upload and prediction interface.
- Documented all experiments with rationale, results, and error analysis in a comprehensive project README.

SentiScan — Sentiment Analysis Classifier | *Python, Hugging Face Transformers, MLflow*

[Live Demo](#) | [GitHub Repo](#)

- Fine-tuned DistilBERT on the IMDb dataset (50K reviews) for binary sentiment classification, achieving 90.6% accuracy and F1 score with a conservative learning rate of 2e-5.
- Conducted controlled experiments comparing learning rates (2e-5 vs. 5e-5), identifying overfitting patterns at epoch 3 and confirming that pretrained Transformer models are sensitive to large LR updates.
- Tracked all training runs with MLflow, logging metrics per epoch for reproducible comparison across experiments.
- Built inference pipeline with FastAPI, containerized with Docker, and deployed to Hugging Face Spaces for live predictions.

Ship It — The Indie Game Reality Coach | *TypeScript, Next.js, Anthropic Claude API, Zod, Render*

[Live Demo](#) | [GitHub Repo](#)

- Built and deployed a full-stack AI web app for the USAII Global AI Hackathon 2026 that converts an over-scoped game idea and the team's real capacity into a buildability analysis, a shippable vertical slice, and a sequenced milestone roadmap.
- Engineered a hybrid architecture: two Anthropic Claude (LLM) calls wrapping a deterministic TypeScript engine, so every estimate comes from auditable code rather than the model, eliminating hallucinated numbers.
- Constrained LLM extraction with Zod-schema structured outputs and validated the effort model against real shipped games, estimating Stardew Valley within ~4% of its reported ~16,000 dev-hours.
- Hardened for production with 70 deterministic tests, per-IP rate limiting, and input-length validation; deployed on Render via an infrastructure-as-code blueprint with automatic deploys.

Skillify — Full-Stack Skills Exchange Platform | *Java, MySQL, JavaScript*

- Developed a credit-based skill exchange web application using Java Servlets and MySQL with a multi-layer backend architecture (DAO pattern, JDBC).
- Designed a normalized relational database, implemented user authentication and session management, and built a responsive frontend with HTML, CSS, and JavaScript.

Technical Skills

ML & Deep Learning: PyTorch, Hugging Face (Transformers, Datasets), scikit-learn, MLflow, Transfer Learning, CNNs, Fine-Tuning

Languages: Python, Java, C++, C, SQL, JavaScript

Deployment & DevOps: FastAPI, Docker, Render, Git, GitHub

Data & Backend: NumPy, pandas, MySQL, JDBC, DAO Pattern, Java Servlets

Core Concepts: Data Structures & Algorithms, OOP, Database Design & Normalization, Computer Networking

Research & Academic Experience

Transformative Sustainability Project | *Stevens Initiative | Fall 2023*

- Participated in a collaborative research project with US-based students on sustainability, resulting in work displayed at the COP 28 UAE Green Zone.

Education

Bachelor of Science in Computer Science

American University of Sharjah | Expected Graduation: 2027

Languages: English (Fluent), Arabic (Fluent)